

HeartRateDataProcessing1

Heart-Rate Data Processing - Part 1

You are a junior data engineer at a Taiwan-based wearable health-tech firm called Seng-Links. Your company has just discovered that the raw heart-rate exports from prototype devices are inconsistent. Some files contains malformed records, impossible values, or missing entries. Before the machine learning team can study usage and identify sleep & exercise patterns, the data engineering team must build a batch-processing pipeline that reads raw files cleans records, computes quality checks, and calculates descriptive statistics for later analysis.

Your company has provided you a folder named "data/", which contains 4 files of heart-rate samples from one participant. The participants device records heart rate every 5 minutes. Your task is to write a Python program that:

1. Reads raw files from the `data/` folder.
2. Cleans the records by ignoring invalid or missing values.
3. Computes and prints descriptive statistics for each file.

Learning Objectives

- Use **Python file I/O** to read in binary data for data processing and use **typecasting** to convert strings into integer data-types.
- Use **for-loops** to calculate the **average** of a Python list and detect malformed data via **conditionals**.
- Document complex chunks of code using **Python comments** and introductory **docstrings**.
- Create **functions** to begin to understand how modules enable separation of responsibilities and integrating separate components to support systems-level thinking.
- **(advanced)** Calculate **rolling-average** without using the **statistics** package in Python. Aim for a computationally efficient solution (no double-nested for-loops).

Instructions

Download this [folder](#) [Download folder](#) to get started with your project.

Begin this project by reading all sections of the **instructions.md** document. Once you understand the directions, utilize documentation, your peers, readings, and classroom notes to complete this project. Some sections of this project have been commented for you, so your task will entail interpreting the comments in order to write the intended line of code. Other sections will test your ability to **independently** implement code with minimal guidance.

Resources

Some relevant internal resources to help you complete this project include the following links:

- **Intro to TLAB 1 (3/12/2026)**

- https://drive.google.com/file/d/1Ew0M_s-dKdJ6-FVnsLA4TwlDunbGDydo/view?usp=sharingLinks to an external site.
- **GitHub Review Session (3/20/2025)**
 - https://theknowledgehouse-org.zoom.us/rec/share/3G7NOqdJGTka4pSCJ0pQU1b1NhV1Js8Nj3_460R9G8glU5Gh1mzHgSMHy_0hARPf.jyWPzZPYwnGHgW6S?startTime=1742505914000Links to an external site.
 - **Passcode:** S#7Hwqcv

Submission

This TLAB will be due 3/20. You have two options to submit your assignment:

1. Submit a link to your Google drive containing your complete code.
2. Submit a link to your GitHub repository containing your completed code

You have 2 attempts for this project. If you submit your project at least 1 week before the due date (3/13) , you will be able to receive an early grade. You can use feedback from this grade to implement any corrections and subsequently resubmit for an opportunity for a higher grade.

HeartRateDataProcessing2

Heart-Rate Data Processing - Part 2

- Submitting a website url

Now that you've completed the first part of your TLAB, you will extend your initial pipeline to reflect how data engineering systems are **structured** and **shared** in real-world environments. Your focus will shift from writing a single script to organizing, maintaining, and distributing your code using GitHub and Python modules.

As you will be iterating on your initial project we will **not** be providing a folder to download for this part of your project. Instead you will continue working on your submitted code in the folder titled "hr_data_pipeline."

Learning Objectives

- Organize code into modular components that separate data ingestion, cleaning, analysis, and output responsibilities.
- Use GitHub to version, manage, and share code, demonstrating an understanding of basic version control workflows.
- (challenge) Generate basic visualizations to explore trends in heart-rate data and support interpretation of results.

Instructions

First, you will modularize your pipeline by separating related functionality into **different files**. For example, data cleaning and statistical calculations. This reinforces how functions support separation of responsibilities and how modules enable more scalable, readable systems.

You will also push your project to GitHub, practicing basic version control workflows. This includes organizing your files, committing changes, and publishing your code. This step reflects how engineers share, track, and collaborate on code in professional environments.

Challenge

The following challenge section is **optional**. You do not need to complete this section to submit your project.

You will implement a **variance** function, which will be used to calculate standard deviation. This builds on your earlier work with averages and introduces a deeper understanding of dispersion, reinforcing how multiple functions can work together to produce more complex metrics. Finally you will also implement a function which takes in a list of integers and creates a matplotlib lineplot image within your repository. This image should be saved to a folder named "images."

Note: If you have incomplete sections in Part 1 of this TLAB, you should prioritize **finishing those foundational components** before moving on to these additional features.

Resources

Some relevant internal resources to help you complete this project include the following links:

- **GitHub Review Session (3/20)**

- https://theknowledgehouse-org.zoom.us/rec/share/3G7NOqdJGTka4pSCJ0pQU1b1NhV1Js8Nj3_460R9G8glU5Gh1mzHgSMHy_0hARPf.jyWPzZPYwnGHgW6S?startTime=1742505914000Links to an external site.
- **Passcode:** S#7Hwqcv

ShoppingDataAnalysis

[TLAB] Shopping Data Analysis - Part 1 (Due April 30)

You are a junior data analyst at **FlashFash**, a fast-growing online shopping company headquartered in Bengaluru, India, that recently went public at a **\$90 Billion USD** IPO valuation. To establish a foothold in the global market, your team has been tasked with analyzing the shopping behavior of American consumers.

Your job will entail analyzing a sample dataset for aggregate descriptive statistics and visualizations. We are interested in discovering the "seasonality" of shoppers, which demographics buy the most, and which market outreach strategies we can apply to maximize purchases.

Download [the starter folder](#) to get started.

Instructions

Every good data analysis starts with curiosity. Before you can answer questions, you need to know which questions are worth asking.

For this first part of your project, you will formulate **6 original questions** to ask about this dataset and write them down in your README.md (that's it!). Check out the instructions listed in `instructions.md` to get started and for more detail.

Resources

- [W5D3: Asking "Good" Questions on Data](#)
- [Asking the right questions of data is a lost artLinks to an external site.](#)

ViewingTimeA/BTest-Part1

Viewing Time A/B Test - Part 1 (Due April 12)

Intro to TLAB #3 - Part 1

You are a data analyst at **NotGPT**, an educational company that creates workshops for professionals to learn data-focused skills.

Your team recently ran a training workshop and applied a treatment to one group of employees while leaving a second group unchanged. The treatment in this A/B test entails a redesign in slide format (the average word count per slide in the original presentation was 50 words, while the average word count per slide in the new presentations is 15 words).

Both groups' viewing times (in minutes) were recorded. Your job is to write a Python pipeline in `explore.ipynb` that cleans, explores, and reports on this data to determine whether the treatment had a meaningful effect on how long employees engaged with the workshop.

This project is organized into three parts, ``cleaner.py`` and ``explore.ipynb``. For each step, You will write a reusable cleaning function, explore the data programmatically, and report your findings in plain language backed by one effective data visualization.

The skills that you will practice in this project include the following:

- File I/O
- Data cleaning
- Descriptive statistics
- Data visualizations
- Creating data reports

Instructions

Download this [folder](#) **Download folder** to get started with your project.

Begin this project by reading all sections of the **instructions.md** document. Once you understand the directions, utilize documentation, your peers, readings, and classroom notes to complete this project. Some sections of this project have been commented for you, so your task will entail interpreting the comments in order to write the intended line of code. Other sections will test your ability to **independently** implement code with minimal guidance.

Resources

Some relevant internal resources to help you complete this project include the following links:

- **GitHub Review Session (3/20/2025)**
 - https://theknowledgehouse-org.zoom.us/rec/share/3G7NOqdJGTka4pSCJ0pQU1b1NhV1Js8Nj3_460R9G8glU5Gh1mzHgSMHy_0hARPf.jyWPzZPYwnGHgW6S?startTime=1742505914000Links to an external site.

- **Passcode:** S#7Hwqcv